TBi-Mamba: Rethinking Joint Classification of Hyperspectral and LiDAR Data With Bidirectional Mamba

Cuiping Shi, *Member, IEEE*, Fei Zhu, Kaijie Shi, Liguo Wang, *Member, IEEE*, and Haizhu Pan, *Member, IEEE*

***Abstract*—Multisource remote sensing image classification based on Mamba has received increasing attention. However, existing methods do not consider the noncausal characteristics of visual data, which leads to insufficient extraction of global features by the model. To alleviate this problem, a novel network called TBi-Mamba is proposed for joint classification of hyperspectral and light detection and ranging (LiDAR) data. First, a cross-modal knowledge search (CMKS) module is designed, which effectively captures local features in different modalities through multiscale feature extraction and interaction between multimodal data. Second, a triple bidirectional sequence scanning mamba module (TBi-M) is proposed, which comprehensively considers multimodal information from the perspective of bidirectional sequence scanning and introduces Mamba to accurately model global dependencies. Finally, a mixed feature reconstruction module (MFRM) is constructed. This module constructs an auxiliary loss function by reconstructing images of different modalities, providing more comprehensive supervision information, and thus improving the performance of the model. The proposed method was evaluated on three publicly available datasets, and the experimental results fully demonstrated that the classification performance of the proposed method is superior to that of some state-of-the-art (SOTA) methods.**

***Index Terms*—Cross-modal, hyperspectral, light detection and ranging (LiDAR), Mamba, reconstruction.**

I. Introduction

HYPERSPECTRAL image (HSI) has shown broad application prospects in agriculture [1], medical [2], military [3], and other fields due to its unique advantages of spectral–spatial integration. However, the HSI data collection process is easily affected by atmospheric and lighting conditions, resulting in reduced classification accuracy. With the development of remote sensing technology, multisensor collaborative observation has become possible [4]. Light detection and ranging (LiDAR) data can accurately obtain elevation information of ground objects and are less affected by atmospheric conditions [5]. Fusion of HSI and LiDAR data can effectively make up for the shortcomings of a single data source and improve the accuracy and reliability of ground object classification [6].

Early classification methods for HSI and LiDAR data fusion mainly focused on using machine learning techniques to extract shallow features inherent in the data [7], [8], [9]. Ge et al. [10] proposed to extract the extinction profile and local binary pattern features from both sources. Then, the derived features of each source are classified by a collaborative representation-based classifier with Tikhonov regularization. Brell et al. [8] used the intensity information of LiDAR data to perform illumination correction on HSI data through ray tracing technology, alleviating problems such as shadows and illumination changes. The graph-based fusion method proposed by Liao et al. [11] achieved the fusion of different feature spaces through a weighted graph structure, improving classification accuracy.

However, these methods overly rely on prior knowledge and have difficulty adaptively learning deep feature representations for data of different modalities. Deep learning-based methods can automatically learn more effective features from input data in an end-to-end manner. Unlike traditional methods, they do not rely on expert experience, which significantly reduces the workload of feature engineering [12], [13], [14], [15], [16].

Convolutional neural network (CNN) and Transformer are representative models in the field of deep learning (DL) [17], [18]. CNNs are good at capturing the essential features of local spatial patterns, yet the issue of inefficient expansion of the receptive field has long existed. Transformer can process sequence data in parallel and capture long-range dependencies. However, Transformer has high demands for computational resources and insufficient sensitivity to the order of input sequences.

Mamba is a novel sequence backbone model that successfully addresses the aforementioned challenges through its innovative state-space model (SSM) architecture [19], [20]. Mamba can efficiently process long-sequence data with linear complexity, which means that it can more effectively capture

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long-range feature correlations across bands and spaces in hyperspectral and LiDAR data without compromising computational efficiency [21], [22].

The causal nature of sequence data mandates that the modeling of the current state strictly depends on an autoregressive generation process of historical states. This unidirectional dependency aligns seamlessly with the temporal characteristics of language. However, in visual data, the semantic attributes of any given pixel are influenced not only by its local neighborhood but also by bidirectional interactions with the global spatial context.

Existing Mamba-based methods for the joint classification of HSI and LiDAR data often overlook the noncausal nature of images, instead adhering to the causal attention mechanisms inherent in sequence modeling. This limitation impedes the models' ability to effectively capture long-range spatial dependencies among pixels and hinders the extraction of global semantic correlations across multimodal remote sensing data.

To alleviate this problem, this article proposes a novel network called TBi-Mamba for the joint classification task of HSI and LiDAR data. First, a cross-modal knowledge search (CMKS) module is designed. Through multiscale feature extraction and interactive guidance between modalities, CMKS can effectively extract unique local features in different modalities. Second, a triple bidirectional sequence scanning mamba module (TBi-M) is proposed. From the perspective of bidirectional sequence scanning, this module utilizes Mamba to model global context information, to more accurately capture long-distance dependencies. Finally, a mixed feature reconstruction module (MFRM) is constructed. This module reconstructs the loss function by reconstructing input images of different modalities, providing more comprehensive supervision information, and further improving the performance of the model. The proposed method has been fully evaluated on three publicly available datasets. The experimental results show that the proposed method has better classification performance than some state-of-the-art (SOTA) methods.

The main contributions of this article are summarized as follows.

1) To enhance the feature extraction of different modal data, a CMKS module is designed. This module effectively captures local features in different modalities through multiscale feature extraction and interaction between multimodal data.
2) To adapt to the noncausal characteristics of visual data, a TBi-M module is proposed, which accurately models global dependencies among diverse feature types through bidirectional sequence scanning.
3) To further enhance model performance, an MFRM module is constructed, which constructs an auxiliary loss function based on the reconstructed images of different modalities, offering richer supervisory information.

The remainder of this article is organized as follows. Section II lists some related works. Section III introduces the proposed TBi-Mamba method in detail. Section IV provides some comparative experiments and a detailed analysis of the experimental results. Finally, some conclusions are provided in Section V.

II. Related Works

A. CNN-Based Methods

CNN, as one of the most representative models in the field of DL, has been widely applied to various computer vision tasks owing to its parameter sharing and local perception mechanisms [23], [24].

Hong et al. [25] proposed a deep encoder–decoder network (EndNet). EndNet reconstructs the multimodal inputs step by step by strengthening fused features, thereby making up for the lack of feature diversity of single-modal data. Fang et al. [26] proposed a spatial–spectral cross-modal enhancement network (S2ENet). S2ENet enhances the information interaction between different modalities and improves their semantic complementarity, thereby achieving more effective feature fusion. Feng et al. [27] proposed a dynamic scale hierarchical fusion network (DSHF). DSHF achieves effective fusion of HSI and LiDAR data by combining dynamic scale feature selection and a multiattention mechanism. Gao et al. [28] proposed an adaptive multiscale spatial–spectral enhancement network (AMSSE-Net). AMSSE-Net effectively alleviates the issues related to feature extraction and fusion in HSI and LiDAR data classification by introducing adaptive multiscale spatial–spectral enhancement and an adaptive feature fusion module.

The convolution operation of CNN is essentially a local operation. This design advantage enables the network to purposefully capture local details in images. However, for scenarios requiring the integration of full-image information, CNN struggles to directly capture dependency relationships within the global scope.

B. Transformer-Based Methods

The self-attention mechanism of Transformer calculates the association weights between any positions in the sequence directly through the triple operation of query–key–value, thereby achieving long-distance semantic alignment [29], [30], [31], [32], [33].

Ding et al. [34] proposed a global–local Transformer network (GLT). GLT uses CNN to learn local correlation features and uses a self-attention mechanism to learn long-range dependencies. In addition, the classification performance is further improved through strategies such as multiscale feature fusion and probabilistic decision fusion. Roy et al. [35] proposed a multimodal fusion Transformer (MFT). MFT adds LiDAR data as classification tokens to the Transformer encoder to help achieve better generalization performance. Zhao et al. [36] proposed a hierarchical CNN and Transformer (HCT). HCT effectively extracts features of multimodal data by combining the advantages of CNN and Transformer, and fuses them through a cross-token attention mechanism. Yao et al. [37] proposed an extended vision Transformer (ExViT). ExViT integrates the spatial and spectral information of multimodal remote sensing images by introducing separation convolution and cross-modal attention mechanism, thus improving the classification performance. Roy et al. [38] proposed a cross-hyperspectral and LiDAR attention Transformer (CrossHL). CrossHL improves the ability to establish long-range

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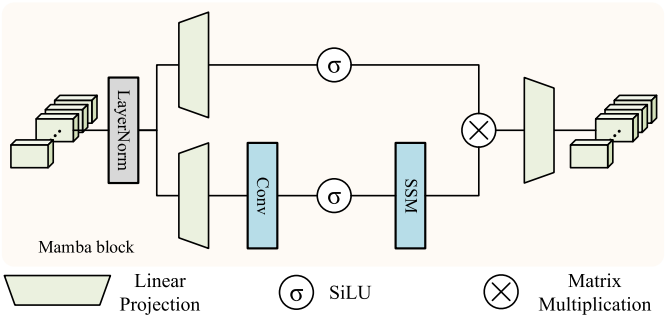


Fig. 1. Structure of Mamba.

relationships by combining the elevation information of LiDAR data with the spectral and position information in HSI data, thereby enhancing the classification accuracy.

Although Transformer performs excellently in capturing long-distance dependencies, its quadratic computational complexity issue is also quite prominent.

C. Mamba-Based Methods

Mamba is a model specifically designed for processing long sequences. It can perceive sequence information globally, and its sequence modeling capability is comparable to that of Transformer, while it has better resource consumption and computational efficiency [39], [40].

Liao et al. [41] proposed a joint classification network of HSI and LiDAR data based on Mamba (HLMamba), which significantly improves the classification performance by effectively fusing HSI and LiDAR data and making full use of edge information. Gao et al. [42] proposed a multiscale feature fusion Mamba (MSFMamba). MSFMamba effectively integrates multiscale spatial features, spectral features, and multisource features, significantly improving the joint classification performance of multisource remote sensing data. Zhang et al. [43] proposed a spatial–spectral cross-Mamba (S^2CrossMamba). By dynamically updating multimodal state information and adopting a multiparameter learning approach to effectively fuse different input modalities, S^2CrossMamba achieves accurate classification of high-dimensional data. He et al. [44] proposed a slice Mamba, which constructs sequences from horizontal and vertical perspectives to facilitate information interaction between pixels, thereby effectively modeling spatial–spectral features. Li et al. [45] proposed a semi-supervised Mamba (Semi-Mamba), which alleviates the problem of lacking large-scale labeled data through Mamba cross-modal fusion and multimodal semi-supervised mechanisms.

III. Methodology

A. Preliminaries

Mamba is an improved version of the SSM, which structure is shown in Fig. 1. Assuming that $x(t) \in \mathbb{R}^N$ is a 1-D input sequence, SSM is a linear time-invariant system that maps $x(t)$ to an output sequence $y(t)$. This process can be expressed as

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \tag{1}$$

$$y(t) = \mathbf{C}h(t). \tag{2}$$

Equations (1) and (2) are the state transfer equation and output equation, respectively. Here, $h(t) \in \mathbb{R}^N$ is the hidden state and $h'(t) \in \mathbb{R}^N$ is the derivative of $h(t)$. $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix, and $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ are the input matrix and output matrix, respectively.

However, SSM is a continuous-time model. To adapt to the discrete data in image processing tasks, the zero-order hold (ZOH) rule is utilized to convert the continuous parameters $(\mathbf{A}, \mathbf{B})$ into discrete parameters $(\bar{\mathbf{A}}, \bar{\mathbf{B}})$. This process can be expressed as

$$\bar{\mathbf{A}} = \exp(\Delta\mathbf{A}) \tag{3}$$

$$\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - I\right) \cdot \Delta\mathbf{B} \tag{4}$$

where Δ is the time scale parameter. After the above discretization process, (1) and (2) can be re-expressed as

$$h_t = \bar{\mathbf{A}}h_t + \bar{\mathbf{B}}x_t \tag{5}$$

$$y_t = \mathbf{C}h_t. \tag{6}$$

In addition, (5) and (6) can be executed in parallel more efficiently using the following formulas:

$$y = x * \bar{\mathbf{K}} \tag{7}$$

$$\bar{\mathbf{K}} = \left(\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}\bar{\mathbf{B}}, \ldots, \mathbf{C}\bar{\mathbf{A}}^t\bar{\mathbf{B}}\right) \tag{8}$$

where * represents the convolution operation and $\bar{\mathbf{K}}$ represents the convolution kernel.

B. Overall Structure

As shown in Fig. 2, the proposed TBi-Mamba mainly consists of four parts: CMKS, TBi-M, MFRM, and classifier. CMKS leverages multiscale feature extraction operators and cross-modal guidance to robustly capture local features from diverse modalities. Subsequently, feature aggregation is employed to transform multiscale features into low-dimensional representations, thereby enhancing feature discriminability. TBi-M comprehensively considers multimodal information from the perspective of bidirectional sequence scanning and introduces Mamba to accurately model global dependencies. MFRM leverages multiple modalities to reconstruct the input image of the target modality and then constructs an auxiliary loss function to provide additional supervision. Finally, through the designed adaptive fusion strategy, the local features are fused with the global features to complete the final classification prediction.

C. CMKS Module

Existing joint classification methods typically employ straightforward strategies like cascading or addition for feature-level fusion, constraining the deep interaction between features from different modalities. To address this limitation, a CMKS module is proposed, as illustrated in Fig. 2.

Assume that HSI inputs are $\mathbf{X}' \in \mathbb{R}^{H\times W\times B}$ and LiDAR inputs are $\mathbf{X}'' \in \mathbb{R}^{H\times W}$, where H and W represent the height and width of the input data, respectively, and B represents the number of bands of HSI input. For the input data of each modality, we adopt three different scales: small scale λ,

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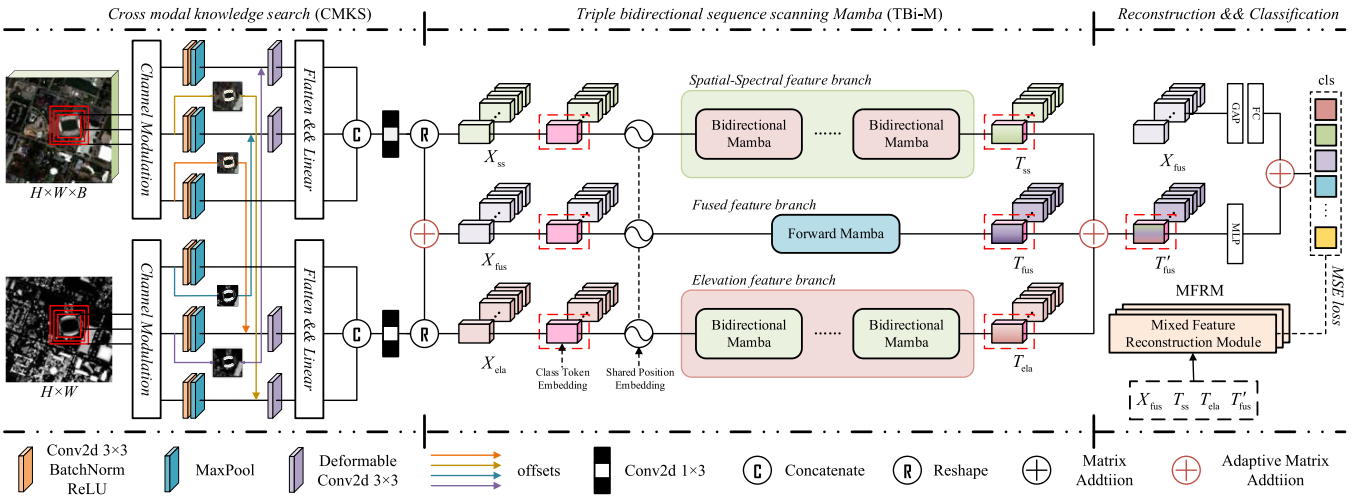


Fig. 2. Overall structure of the proposed TBi-Mamba. The CMKS on the left extracts local features from diverse modalities and generates more discriminative feature representations. The TBi-M in the middle utilizes multiple scanning strategies to capture global contextual information in spatial–spectral features, fusion features, and elevation features. Finally, the local and global features are fused to complete the classification prediction, while the image reconstructed by MFRM guides the network training process.

medium scale 2λ, and large scale 4λ. Therefore, the input of HSI and LiDAR data can be represented as $\mathbf{X}'_k \in \mathbb{R}^{k\lambda\times k\lambda\times B}$ and $\mathbf{X}''_k \in \mathbb{R}^{k\lambda\times k\lambda}$, respectively.

Among them, $k \in \{1, 2, 4\}$. Considering the significant dimensional difference between HSI and LiDAR data, we initially unify the dimensions of the two modalities through channel modulation to avoid adverse effects on subsequent processing. Subsequently, a 3×3 convolutional block is employed to extract shallow features from these feature maps. Thereafter, maximum pooling is applied to capture high-frequency information within the convolved results. This process can be represented as

$$\mathbf{U}_k = \text{Maxpool}\left(F^*_{3\times3}\left(F_{\text{cm}}\left(\mathbf{X}'_k\right)\right)\right) \tag{9}$$

$$\mathbf{Q}_k = \text{Maxpool}\left(F^*_{3\times3}\left(F_{\text{cm}}\left(\mathbf{X}''_k\right)\right)\right) \tag{10}$$

where $\mathbf{U}_k \in \mathbb{R}^{k\lambda/2\times k\lambda/2\times d}$ and $\mathbf{Q}_k \in \mathbb{R}^{k\lambda/2\times k\lambda/2\times d}$ represent the output feature maps of HSI and LiDAR data, respectively; d represents the dimension after channel modulation; $F^*_{3\times3}$ represents the convolution block, which includes: a convolution layer of size 3×3, a batch normalization layer, and a ReLU activation function; Maxpool represents the maximum pooling; and F_{cm} represents the channel modulation operation.

In the following, the feature map's dimensions are reduced by half. At this stage, the medium- and large-scale feature maps correspond to the original small- and medium-scale feature maps, respectively. Subsequently, cross-modal interactions are established between the two modalities. Using LiDAR data as an example, we extract offsets from the initial small- and medium-scale feature maps of this modality. These offsets are then employed to guide the feature extraction of medium- and large-scale outputs on HSI data, thereby enhancing cross-modal interaction. This process can be represented as

$$O'_i, O''_i \leftarrow \text{Offsets}\left(F_{\text{cm}}\left(\mathbf{X}'_i\right)\right), \text{Offsets}\left(F_{\text{cm}}\left(\mathbf{X}''_i\right)\right) \tag{11}$$

$$\mathbf{U}'_j, \mathbf{Q}'_j \leftarrow \text{DConv}\left(\mathbf{U}_j, O''_i\right), \text{DConv}\left(\mathbf{Q}_j, O'_i\right) \tag{12}$$

$$\mathbf{U}'_1, \mathbf{Q}'_1 \leftarrow \mathbf{U}_1, \mathbf{Q}_1 \tag{13}$$

where O'_i and O''_i represent the output offsets of HSI and LiDAR data, $i \in \{1, 2\}$, respectively; $\mathbf{U}'_j \in \mathbb{R}^{j\lambda/2\times j\lambda/2\times d}$ and $\mathbf{Q}'_j \in \mathbb{R}^{j\lambda/2\times j\lambda/2\times d}$ represent the output feature maps of HSI and LiDAR data, $j \in \{1, 2, 4\}$, respectively; Offsets represents the function that generates the offset; and DConv represents the deformable convolution layer.

Following cross-modal knowledge interaction, feature maps of three different scales are obtained. These feature maps are then flattened into 1-D vectors and linearly mapped to generate sequences of equal length. These sequences are then stacked along a new dimension to form a 2-D tensor. To capture the relationships between features of different scales, a 1×3 strip convolutional kernel is designed. This kernel enables the simultaneous consideration of three-scale features during the sliding window operation. Through this process, multiscale features are aggregated and compressed into a low-dimensional output space, thereby obtaining a feature representation containing richer discriminative information. This process can be represented as

$$\mathbf{L}'_j, \mathbf{L}''_j \leftarrow \text{Linear}\left(\text{Flatten}\left(\mathbf{U}'_j\right)\right), \text{Linear}\left(\text{Flatten}\left(\mathbf{Q}'_j\right)\right) \tag{14}$$

$$\mathbf{H}, \mathbf{E} \leftarrow F^*_{1\times3}\left(\text{Concat}\left(\mathbf{L}'_j\right)\right), F^*_{1\times3}\left(\text{Concat}\left(\mathbf{L}''_j\right)\right) \tag{15}$$

where $\mathbf{L}'_j$ and $\mathbf{L}''_j$ represent the sequences of HSI and LiDAR data, respectively; Linear represents the linear mapping; Flatten represents the flattening operation; $\mathbf{H} \in \mathbb{R}^{l\times d}$ and $\mathbf{E} \in \mathbb{R}^{l\times d}$ represent the output sequences of HSI and LiDAR data, respectively; l represents the length of the sequence; Concat represents the cascade operation; and $F^*_{1\times3}$ represents a strip convolution block, which includes: a convolution layer of size 1×3, a batch normalization layer, and a sigmoid activation function.

D. TBi-M Module

Mamba performs well in multisource joint classification tasks, but it focuses more on causal sequences. However, there are complex local and global dependencies between pixels in image data. It is difficult for unidirectional Mamba to fully

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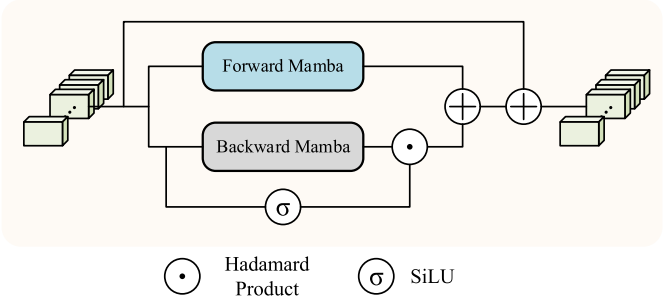


Fig. 3. Structure of bidirectional Mamba.

capture these dependencies. To alleviate this problem, a TBi-M is designed, as illustrated in Fig. 2.

TBi-M employs a multibranch feature extraction structure to process spatial–spectral, fusion, and elevation features independently. Fusion features are derived through the adaptive fusion of diverse modal features. This process can be expressed as

$$\mathbf{V} = (1-\eta)\cdot\mathbf{H} + \eta\cdot\mathbf{E} \tag{16}$$

where $\mathbf{V}$ denotes the fusion feature. Subsequently, classification tokens are added to each feature, and shared positional encoding is applied to incorporate positional information. The results are denoted as $\mathbf{H}, \mathbf{V}, \mathbf{E} \in \mathbb{R}^{(d+1)\times l}$.

In the spatial–spectral feature branch and the elevation feature branch, the bidirectional Mamba (Fig. 3) is employed as the basic unit. Specifically, the forward Mamba adopts a forward sequence scanning strategy, while the backward Mamba employs a backward scanning strategy starting from the sequence's end. In addition, to enhance the representational ability of the model, a SiLU activation function is introduced to perform a nonlinear transformation on the output of the backward Mamba.

Finally, the original sequence, the output of the forward Mamba, and the output of the backward Mamba are fused. In the fusion feature branch, a separate forward Mamba is adopted. This process can be expressed as

$$\begin{cases} \mathbf{T}_{\mathrm{H}} = \mathrm{FMamba}(\mathbf{H}) + \mathrm{BMamba}(\mathbf{H})\cdot\sigma(\mathbf{H}) + \mathbf{H} \\ \mathbf{T}_{\mathrm{V}} = \mathrm{FMamba}(\mathbf{V}) \\ \mathbf{T}_{\mathrm{E}} = \mathrm{FMamba}(\mathbf{E}) + \mathrm{BMamba}(\mathbf{E})\cdot\sigma(\mathbf{E}) + \mathbf{E} \end{cases} \tag{17}$$

where $\mathbf{T}_{\mathrm{H}}, \mathbf{T}_{\mathrm{V}}, \mathbf{T}_{\mathrm{E}} \in \mathbb{R}^{(d+1)\times l}$ represent the output tokens of spatial–spectral features, fusion features, and elevation features, respectively. Subsequently, the classification tokens corresponding to the three features are extracted, namely $\mathbf{T}_{\mathrm{H}}^{\mathrm{cls}}, \mathbf{T}_{\mathrm{V}}^{\mathrm{cls}}, \mathbf{T}_{\mathrm{E}}^{\mathrm{cls}} \in \mathbb{R}^{l}$. To balance the importance of different modal features, $\mathbf{T}_{\mathrm{H}}^{\mathrm{cls}}$ and $\mathbf{T}_{\mathrm{E}}^{\mathrm{cls}}$ are adaptively scaled, and then, the results are fused with $\mathbf{T}_{\mathrm{V}}^{\mathrm{cls}}$ for the final classification prediction. This process can be expressed as

$$\mathbf{T}^{\mathrm{cls}} = \mathbf{T}_{\mathrm{V}}^{\mathrm{cls}} + (1-\eta)\cdot\mathbf{T}_{\mathrm{H}}^{\mathrm{cls}} + \eta\cdot\mathbf{T}_{\mathrm{E}}^{\mathrm{cls}}. \tag{18}$$

E. MFRM Module

To optimize the model training process, an MFRM is introduced. The detailed structure of MFRM is shown in Fig. 4. The specific data used in the reconstruction process depend on the input modality of the reconstruction.

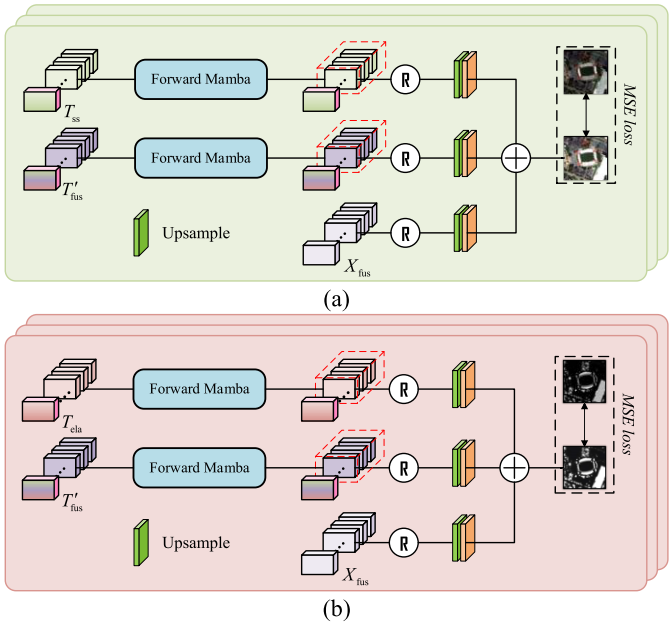


Fig. 4. Structure of MFRM for (a) reconstructing HSI data and (b) reconstructing LiDAR data.

Taking the process of reconstructing HSI input (Fig. 4) as an example, three features are selected, including: spatial–spectral feature $\mathbf{T}_{\mathrm{H}}$, fusion feature $\mathbf{T}_{\mathrm{V}}$, and initial fusion feature $\mathbf{V}$. First, $\mathbf{T}_{\mathrm{H}}$ and $\mathbf{T}_{\mathrm{V}}$ are processed through a forward Mamba transformation to extract feature tokens. These extracted tokens are then dimensionally adjusted to align with the dimensions of $\mathbf{V}$. This process can be expressed as

$$\begin{matrix}\mathbf{T}'_{\mathrm{H}} \\ \mathbf{T}'_{\mathrm{E}} \\ \mathbf{T}'_{\mathrm{V}}\end{matrix} \leftarrow \begin{cases}\mathrm{FMamba}(\mathbf{T}_{\mathrm{H}}) \\ \mathrm{FMamba}(\mathbf{T}_{\mathrm{E}}) \\ \mathrm{FMamba}(\mathbf{T}_{\mathrm{V}})\end{cases} \tag{19}$$

$$\begin{cases}\mathbf{T}''_{\mathrm{H}}, \mathbf{T}''_{\mathrm{E}} \leftarrow F_{\mathrm{re}}\left(\mathbf{T}'_{\mathrm{H}}[:,1:,:]\right), & F_{\mathrm{re}}\left(\mathbf{T}'_{\mathrm{E}}[:,1:,:]\right) \\ \mathbf{T}''_{\mathrm{V}}, \mathbf{V}' \leftarrow F_{\mathrm{re}}\left(\mathbf{T}'_{\mathrm{V}}[:,1:,:]\right), & F_{\mathrm{re}}(\mathbf{V}).\end{cases} \tag{20}$$

In the following, the three feature sets are upsampled to match the scale of the target reconstructed image. Finally, a convolutional block is utilized to fuse the upsampled features, resulting in the reconstructed HSI. This process can be expressed as

$$\begin{matrix}\mathcal{G}_k^{\mathrm{H}} & \mathcal{G}_k^{\mathrm{E}} \\ \mathcal{G}_k^{\mathrm{V1}} & \mathcal{G}_k^{\mathrm{V2}}\end{matrix} \leftarrow \begin{cases}F_k^{\mathrm{up}}(\mathbf{T}''_{\mathrm{H}}), & F_k^{\mathrm{up}}(\mathbf{T}''_{\mathrm{H}}) \\ F_k^{\mathrm{up}}(\mathbf{T}''_{\mathrm{V}}), & F_k^{\mathrm{up}}(\mathbf{V}')\end{cases} \tag{21}$$

$$\begin{cases}\mathcal{G}_k^{\mathrm{hsi}} = F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{H}}\right) + F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{V1}}\right) + F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{V2}}\right) \\ \mathcal{G}_k^{\mathrm{ela}} = F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{E}}\right) + F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{V1}}\right) + F_{3\times3}^{*}\left(\mathcal{G}_k^{\mathrm{V2}}\right)\end{cases} \tag{22}$$

where F_{re} represents the reshaping operation; F_k^{up} represents the upsampling according to the scale of the target image; and $\mathcal{G}_k^{\mathrm{hsi}}$ and $\mathcal{G}_k^{\mathrm{ela}}$ represent the reconstructed HSI and LiDAR data, respectively, $k \in \{1, 2, 4\}$.

During the reconstruction process, the features $\mathbf{T}_{\mathrm{V}}$ and $\mathbf{V}$ are adopted to supplement the global features and local

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Fig. 5. Pseudo-color maps and LiDAR-based DSMs on the Houston 2013 dataset.

features shared by the two modal features, respectively. The spatial–spectral features are utilized to supplement the modal features unique to the HSI. The reconstruction process for LiDAR data follows a similar approach.

F. Construction of Loss Function

The proposed method employs a two-stage training strategy: classification and reconstruction. For the classification task (Fig. 2), after extracting global features, the initial fusion features $\mathbf{V}$ are fused with the final classification feature vector $\mathbf{T}^{\text{cls}}$. The fused features are then input into the classifier, and the cross-entropy loss between the predicted and true labels is computed. This process can be expressed as

$$\mathbf{P} = (1-\eta)\cdot \text{Linear}\left(\text{Avgpool}\left(\mathbf{V}\right)\right) + \eta \cdot \text{MLP}\left(\mathbf{T}^{\text{cls}}\right) \quad (23)$$

$$\hat{Y} = \text{argmax}\left(\mathbf{P}\left[\text{i}\right]\right), \quad i \in [1, \theta] \quad (24)$$

$$\mathcal{L}_{\text{cls}} = -\sum_{i=1}^{\theta} Y_i \log\left(\hat{Y}_i\right) \quad (25)$$

where $\mathbf{P}$ is a list containing the predicted probabilities of all categories, $\mathcal{L}_{\text{cls}}$ represents the classification loss, θ represents the number of categories, $\hat{Y}$ represents the final predicted category, and Y represents the true category.

For the reconstruction task, mean squared error (mse) is adopted as the loss function to measure the difference between the reconstructed image and the original image. Finally, the proposed method is trained end-to-end with the following loss function:

$$\mathcal{L}_{\text{rel}} = \frac{1}{M}\frac{1}{J}\sum_{i=1}^{M}\sum_{j=1}^{J}\left(\left(\mathcal{G}_{i,j,k}^{\text{hsi}} - \mathbf{X}'_{i,j,k}\right)^2 + \left(\mathcal{G}_{i,j,k}^{\text{ela}} - \mathbf{X}''_{i,j,k}\right)^2\right) \quad (26)$$

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{rel}} \quad (27)$$

where $\mathcal{L}_{\text{rel}}$ represents the reconstruction loss, $\mathcal{L}$ represents the overall loss, M represents the number of different scales, and J represents the number of reconstructed samples. $\mathcal{G}_{i,j,k}^{\text{hsi}}$ represents the jth reconstructed sample in the reconstructed HSI input at scale k. $\mathbf{X}'_{i,j,k}$ represents the jth real sample in the HSI input at scale k.

IV. Experimental Results and Analysis

A. Dataset Description

1) Houston 2013 [46]: This dataset was collected by the National Center for Airborne Laser Mapping in June 2012 at the University of Houston and its surrounding areas. The dataset contains two types of data: HSI data and LiDAR data (Fig. 5). The HSI data are acquired by the Compact Airborne Spectrographic Imager (CASI) sensor, covering the visible and near-infrared bands (380–1050 nm) with a spatial resolution of 2.5 m; the LiDAR data have the same spatial resolution and size as the HSI data but only contain one band, namely elevation information. The size of the entire dataset is 349 × 1905 pixels, and it contains a total of 15029 labeled samples, covering 15 types of ground objects. The training and testing samples used in classification tasks are presented in Table I.

TABLE I
Training and Test Sample Numbers for the Houston2013 Dataset

Class Name	Train	Test	Class Name	Train	Test
Healthy Grass	20	1231	Road	20	1232
Stressed Grass	20	1234	Highway	20	1207
Synthetic Grass	20	677	Railway	20	1215
Trees	20	1224	Parking Lot 1	20	1213
Soil	20	1222	Parking Lot 2	20	449
Water	20	305	Tennis Court	20	408
Residential	20	1248	Running Track	20	640
Commercial	20	1224			
Total				300	14729



Fig. 6. Pseudo-color maps and LiDAR-based DSMs on the Trento dataset.

TABLE II
Training and Test Sample Numbers for the Trento Dataset

Class Name	Train	Test	Class Name	Train	Test
Apple Trees	20	4014	Woods	20	9103
Buildings	20	2883	Vineyard	20	10481
Ground	20	459	Roads	20	3154
Total				120	30094

2) Trento [47]: This dataset was collected in a rural area south of Trento, Italy. The dataset contains two types of data: HSI data and LiDAR data (Fig. 6). HSI data are acquired by the Airborne Hyperspectral Imaging System Eagle sensor, covering the visible and near-infrared bands (0.42–0.99 μm), with 63 spectral bands. LiDAR data are acquired by the Optech Airborne Laser Topography 3100EA sensor, which has the same resolution as HSI data and can accurately describe surface elevation. The size of the entire dataset is 166 × 600 pixels, containing a total of 30214 labeled samples, covering six ground object categories. The training and testing samples used in classification tasks are presented in Table II.

3) MUUFL [48], [49]: This dataset was collected on the campus of the University of Southern Mississippi Gulf Park in November 2010. The dataset contains two types of data:

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Fig. 7. Pseudo-color maps and LiDAR-based DSMs on the MUUFL dataset.

TABLE III
TRAINING AND TEST SAMPLE NUMBERS FOR THE MUUFL DATASET

Class Name	Train	Test	Class Name	Train	Test
Trees	20	23226	Buildings Shadow	20	2213
Mostly Grass	20	4250	Buildings	20	6220
Mixed Ground Surface	20	6862	Sidewalk	20	1365
Dirt and Sand	20	1806	Yellow Curb	20	163
Road	20	6667	Cloth Panels	20	249
Water	20	446			
Total				220	53467

HSI data and LiDAR data (Fig. 7). HSI data are acquired by the CASI sensor, which originally covers 72 spectral bands (0.38–1.05 μm), but due to noise problems, 64 bands are employed. LiDAR data contain two bands: digital elevation model (DEM) and laser pulse intensity image. DEM provides elevation information, and the laser pulse intensity image reflects the surface reflection intensity. The size of the entire dataset is 325×220 pixels, containing a total of 53687 samples, covering 11 ground object categories. The training and testing samples used in classification tasks are presented in Table III.

B. Experimental Configuration

1) Hardware Configuration: All experiments in this article were conducted on the 2.1.2 version of the PyTorch framework. To ensure the efficiency of the experiments, the experimental environment was configured as follows: Intel Core i9-14900KF CPU (128-GB RAM) and NVIDIA GeForce RTX 4090 GPU (24-GB VRAM).

2) Evaluation Indicators: To comprehensively evaluate the performance of each classification method, this article employs three common indicators, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ). In addition, to improve the reliability of the experimental results, all experimental results are the average of ten independent experiments.

3) Parameter Analysis: To objectively evaluate the performance of all methods, the comparison methods all adopted the optimal hyperparameter settings in their papers. The proposed method employs the Adam optimizer to train the network, and the training epoch is set to 500. The learning rate and batch size are critical hyperparameters for network training. A series of experiments were conducted, with the learning rate varying from $1e^{-4}$ to $5e^{-3}$ and the batch size from 16 to 128 while keeping the learning rate decay coefficient fixed at 0.9. As depicted in Fig. 8, the experimental results indicate that the optimal performance for the Houston 2013 and MUUFL datasets is achieved with a batch size of 64 and a learning rate of $5e^{-4}$. For the Trento dataset, the best results are obtained with a learning rate of $5e^{-4}$ and a smaller batch size of 16. In addition, the spatial size of the multiscale input is set to $\lambda \times \lambda$, $2\lambda \times 2\lambda$, and $4\lambda \times 4\lambda$. The optimal settings of λ on the three datasets are shown in Fig. 9. Specifically, for the Houston 2013 dataset, when $\lambda = 8$, the model achieves the best classification performance. For the Trento dataset, the optimal setting of λ is 6. For the MUUFL dataset, the optimal setting of λ is 4.

TABLE IV
ABLATION EXPERIMENTS

Case	Components			Datasets		
	CMKS	TBi-M	MFRM	Houston	Trento	MUUFL
1	√	-	-	95.57	98.99	84.79
2	√	√	-	96.13	99.42	85.41
3	√	-	√	96.39	99.32	85.54
4	√	√	√	**96.46**	**99.63**	**86.07**

C. Ablation Analysis of CMKS

The proposed CMKS module extracts local features of different modal data through multiscale feature extraction and cross-modal interaction. In this section, the effectiveness of the CMKS module is evaluated through some ablation experiments. The experimental results are shown in Fig. 10. It can be seen that multiscale structure can achieve better classification performance than single-scale structure, indicating that multiscale features help improve the representation ability of the model; However, the classification performance of the complete CMKS module is significantly better than model that only use multiscale structure, fully demonstrating the important role of cross-modal interaction in local feature extraction. These experimental results fully prove the effectiveness of the CMKS module.

D. Ablation Analysis of TBi-M and MFRM

To verify the effectiveness of the proposed TBi-M and MFRM modules, four sets of ablation experiments were conducted on three datasets: Houston 2013, Trento, and MUUFL, using OA as the evaluation indicator. Here, the CMKS module is used as the baseline, and the performance differences are compared by removing different modules. The experimental results are shown in Table IV.

1) Ablation Analysis of TBi-M: For the proposed TBi-M module, the noncausal characteristics of image are fully considered through bidirectional sequential scanning. As shown in Case 1 of Table IV, the baseline, which solely uses the CMKS module, exhibits the lowest classification performance. By incorporating the TBi-M module into the baseline in Case 2, a significant improvement in classification accuracy is obtained. Specifically, the accuracy on the three datasets increases

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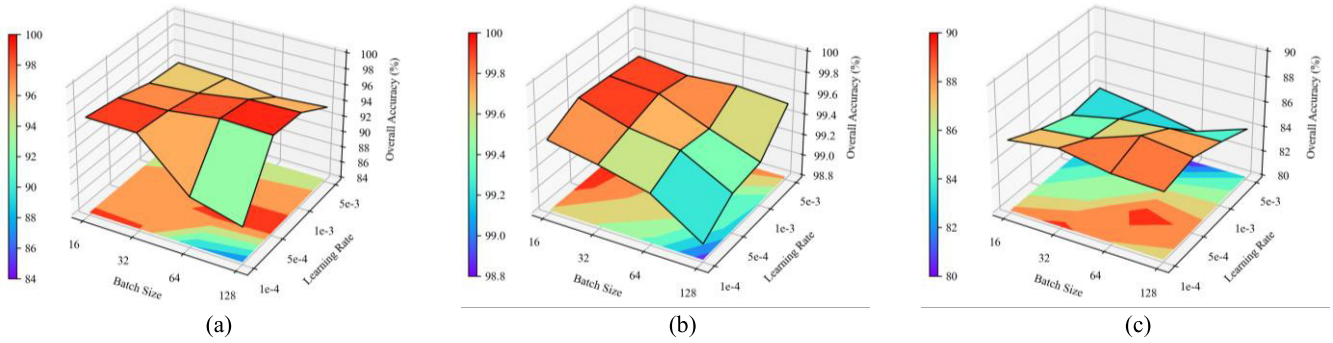


Fig. 8. Impact of learning rate and batch size on performance. (a)–(c) Houston 2013, Trento, and MUUFL, respectively.

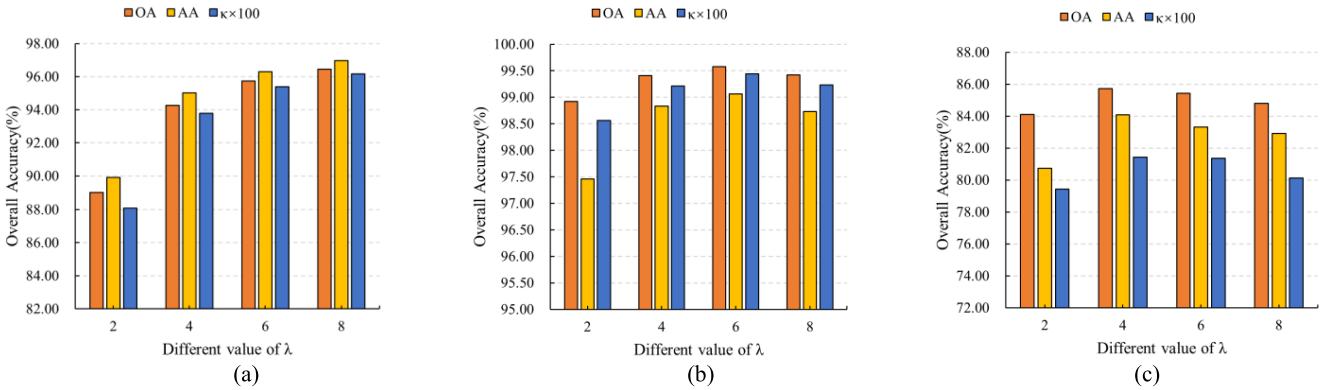


Fig. 9. Impact of different spatial sizes on performance. (a)–(c) Houston 2013, Trento, and MUUFL, respectively.

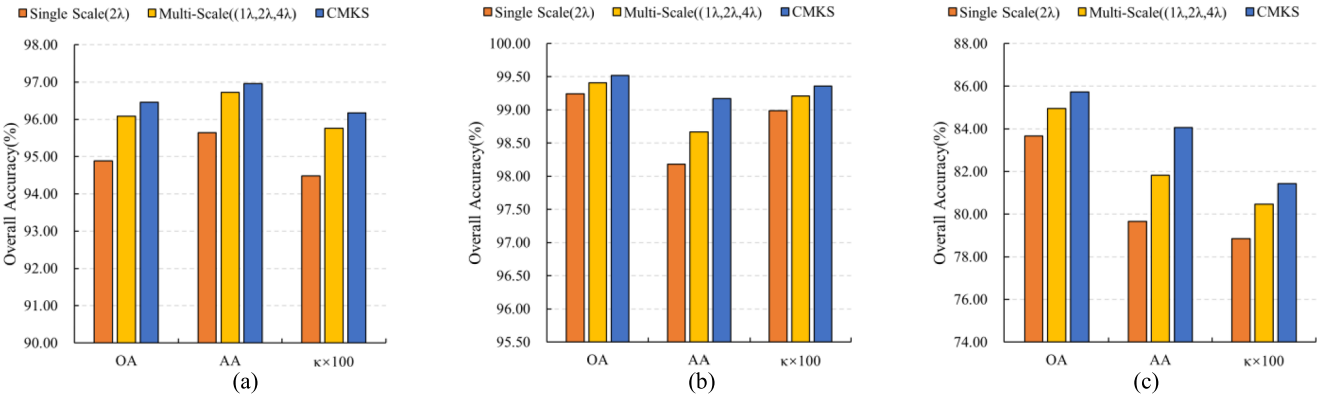


Fig. 10. Ablation analysis of CMKS. (a)–(c) Houston 2013, Trento, and MUUFL, respectively.

by 0.56%, 0.43%, and 0.62%, which fully demostrates the effectiveness of the bidirectional sequence scanning for global dependency modeling.

2) Ablation Analysis of MFRM: For the proposed MFRM module, input data of different modalities are reconstructed by combining multiple features and utilized to guide the training process of the model. In Case 3 of Table IV, the integration of the MFRM module with the baseline network achieves a significant improvement in classification performance. Compared to the baseline, the accuracy on the three datasets increases by 0.82%, 0.33%, and 0.75%.

Furthermore, Case 4 achieves the optimal classification performance by incorporating all proposed components. This result highlights the effective collaboration between the CMKS, TBi-M, and MFRM modules.

E. Different Scanning Strategies

To demonstrate the effectiveness of the proposed method integrating the bidirectional scanning strategy into joint classification, a detailed analysis of the applicable scanning strategies for the Houston, Trento, and MUUFL datasets was conducted in this section, and the specific results are summarized in Table V. Cases 1–7 represent different bidirectional scanning strategies, namely Row, Col, Zigzag, and Col&Zigzag (Fig. 11).

Specifically, when the sequence flattening is performed in a row-first [Fig. 11(a)] manner, the proposed method achieves the best performance in the Houston scene. When the column-first [Fig. 11(b)] flattening sequence is adopted, the proposed method exhibits the optimal performance in the Trento scene. In the MUUFL scene, the proposed method

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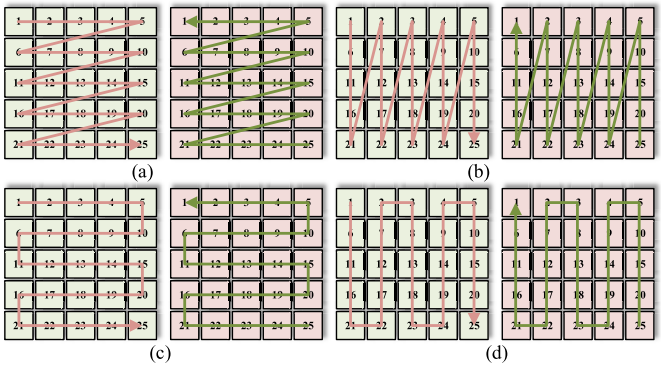


Fig. 11. Different scanning strategies. (a)–(d) Row, Col, Zigzag, and Col&Zigzag.

TABLE V
DIFFERENT SCANNING STRATEGIES

Case	Scanning Strategy	Houston 2013	Trento	MUUFL
1	(a)	**96.46**	99.52	85.73
2	(b)	96.15	**99.63**	85.60
3	(c)	96.23	99.57	85.84
4	(d)	96.27	99.53	**86.07**
5	(a)+(b)	96.18	99.45	85.87
6	(a)+(c)	96.24	99.48	85.82
7	(a)+(d)	96.01	99.35	85.84

attains the best performance when the column-first Zigzag [Fig. 11(d)] approach is utilized.

Furthermore, the bidirectional scanning was extended to four-way scanning by combining Case 1 with other scanning strategies. However, the experimental results of Cases 5–7 indicate that the performance of the model using four-way scanning is consistently inferior to that of bidirectional scanning. This suggests that the bidirectional scanning strategy strikes a better balance between computational efficiency and classification accuracy, and the additional scanning directions in the four-way scanning may introduce redundant information or increase computational complexity without significantly improving the classification performance, thus highlighting the superiority of the bidirectional scanning strategy employed in the proposed method.

F. Comparison Methods

To comprehensively evaluate the performance of the proposed TBi-Mamba method, we compare it with several SOTA methods. These methods are briefly described as follows.

1) *EndNet [24]:* This network is a pixel-to-pixel multimodal classification method, which mainly consists of three parts: the feature extraction network FE-Net, the fusion network F-Net that integrates features of different modalities, and the image reconstruction network R-Net.
2) *S2ENet [25]:* Compared with traditional methods, this method employs the spatial–spectral enhancement module (S2EM) to enhance the spatial representation of HSI and the spectral representation of LiDAR, effectively promoting the interaction and understanding between multisource data.
3) *GLT [34]:* This network extracts multiscale local spatial features through CNN and uses Transformer to capture global spectral features. Then, an adaptive fusion strategy is used to effectively integrate these global-local features, thereby improving the classification accuracy of the model.
4) *MFT [35]:* The MFT extracts features from HSI data using CNN. Subsequently, the LiDAR data are converted into classification tokens and concatenated with multiple feature tokens extracted from HSI. Finally, the interaction and fusion of different modal features is achieved through a multihead self-attention mechanism.
5) *HCT [36]:* The HCT extracts features from HSI and LiDAR data through CNN. Then, the extracted features are tokenized and fused through a cross-modal interactive encoder. Finally, an MLP is employed to complete the classification prediction.
6) *ExViT [35]:* In this method, the depthwise separable convolution is introduced into ViT, which can better capture the spatial and channel features of the image. In addition, the introduction of the cross-modal attention mechanism promotes the interaction and fusion of different modal features.
7) *DSHF [37]:* The DSHF uses the dynamic scale feature extraction (DSFE) module to adaptively fuse features of different scales to improve the efficiency of feature extraction and uses the multiattention fusion module to fuse HSI and LiDAR features from the shallow and deep layers, respectively, to capture the interactive relationship between them.
8) *CrossHL [38]:* It is an end-to-end multimodal data classification method. The framework extracts deep features of HSI data through CNN and uses the Cross-HL attention mechanism to achieve efficient fusion of HSI and LiDAR data.
9) *S^2 CrossMamba [41]:* It is a multimodal fusion network based on the state-space model. It adopts the inverted bottleneck cross-Mamba structure and the spectral–spatial attention mechanism branches to enhance the feature expression ability.
10) *HLMamba [43]:* This method first extracts edge features from LiDAR data and then fuses HSI, LiDAR data, and edge features through a multimodal feature extraction module. Finally, a multimodal fusion module based on Mamba is used to achieve efficient fusion of multimodal data.

G. Quantitative Experiments

Tables VI–VIII show the quantitative experimental results of the proposed method and the comparison methods on three datasets.

Although EndNet improves the expressiveness of the model by fusing multimodal features and reconstructing images, it is insufficient in mining the spatial features in HSI and LiDAR data, resulting in poor performance on all three datasets. S2ENet effectively promotes information fusion between

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TABLE VI
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE HOUSTON2013 DATASET (OPTIMAL RESULTS ARE BOLDED)

No.	EndNet	S2ENet	GLT	MFT	HCT	ExViT	DSHF	CrossHL	S^2CrossMamba	HLMamba	Proposed
C1	94.13	**95.68**	91.89	92.09	95.26	91.94	90.98	95.09	93.01	94.11	92.73
C2	96.49	94.62	98.03	93.23	98.14	98.28	98.54	**98.88**	96.73	98.10	98.28
C3	99.73	99.11	98.73	**99.76**	99.79	98.79	99.26	99.41	98.99	99.55	99.59
C4	94.97	96.52	97.59	96.69	97.81	97.17	98.52	98.55	96.21	98.12	**99.31**
C5	97.76	99.80	99.47	97.07	99.77	96.15	99.83	99.61	99.59	**99.90**	99.84
C6	93.70	97.70	98.22	98.82	98.49	98.62	**99.01**	97.70	96.78	98.55	97.90
C7	88.65	96.27	96.49	91.63	96.68	96.76	95.19	93.75	97.75	**97.99**	97.64
C8	81.54	87.79	87.45	76.86	89.85	86.48	82.59	87.61	**92.47**	89.93	86.55
C9	75.50	90.96	90.32	88.08	88.39	90.25	78.24	86.12	89.64	93.13	**96.19**
C10	81.21	95.74	91.83	88.21	91.58	90.68	93.37	91.79	86.79	93.02	**96.01**
C11	86.97	92.39	93.30	90.07	93.79	95.96	92.18	90.07	95.07	95.67	**97.55**
C12	70.30	91.66	91.78	86.68	91.73	87.27	**96.53**	90.87	88.77	92.46	94.89
C13	62.80	95.10	96.49	96.17	97.23	96.26	95.99	97.19	90.95	97.28	**97.64**
C14	99.17	99.95	99.95	99.46	**100**	99.80	99.75	99.95	99.80	**100**	**100**
C15	98.44	99.84	**100**	99.16	**100**	99.91	99.84	**100**	99.97	**100**	**100**
OA(%)	87.64	94.89	94.70	91.54	95.14	94.06	93.65	94.22	94.30	95.92	**96.46**
AA(%)	88.09	95.54	95.52	92.93	95.90	94.95	94.65	95.10	94.83	96.52	**96.94**
$\kappa(\times 100)$	86.63	94.47	94.27	90.85	94.75	93.58	93.13	93.74	93.83	95.59	**96.17**

TABLE VII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE TRENTO DATASET (OPTIMAL RESULTS ARE BOLDED)

No.	EndNet	S2ENet	GLT	MFT	HCT	ExViT	DSHF	CrossHL	S^2CrossMamba	HLMamba	Proposed
C1	64.37	99.62	99.55	98.28	98.46	98.41	**99.88**	99.64	97.39	99.68	99.85
C2	94.57	98.45	**98.82**	97.29	96.18	97.62	98.04	97.46	97.63	97.83	98.29
C3	93.68	98.30	96.73	95.99	97.65	97.91	98.25	98.39	96.73	98.25	**97.78**
C4	98.88	99.99	**100**	99.99	99.98	**100**	99.90	**100**	**100**	**100**	**100**
C5	78.20	99.99	**100**	99.73	99.71	99.94	99.94	99.93	99.96	99.98	**100**
C6	86.62	93.83	95.59	93.61	95.03	97.30	95.93	92.40	94.95	96.57	**98.56**
OA(%)	85.30	99.12	99.31	98.69	98.76	99.22	99.29	98.86	98.83	99.35	**99.63**
AA(%)	86.06	98.36	98.44	97.48	97.83	98.53	98.66	97.97	97.78	98.72	**99.08**
$\kappa(\times 100)$	80.58	98.83	99.08	98.24	98.35	98.96	99.06	98.49	98.44	99.14	**99.51**

TABLE VIII
CLASSIFICATION PERFORMANCE OBTAINED BY DIFFERENT METHODS FOR THE MUUFL DATASET (OPTIMAL RESULTS ARE BOLDED)

No.	EndNet	S2ENet	GLT	MFT	HCT	ExViT	DSHF	CrossHL	S^2CrossMamba	HLMamba	Proposed
C1	85.91	91.51	89.98	87.87	88.20	91.85	84.88	91.68	90.58	**95.18**	90.50
C2	73.00	74.99	77.79	71.89	75.84	69.47	75.84	76.76	72.29	74.02	**81.14**
C3	69.15	**71.84**	68.49	58.67	65.26	69.51	73.34	66.74	64.32	61.25	70.14
C4	77.59	88.52	85.14	83.12	**91.37**	86.81	89.99	84.74	90.59	85.27	87.41
C5	84.55	79.03	79.87	78.22	71.22	78.77	**91.42**	79.49	72.16	73.24	84.39
C6	98.92	98.83	99.55	99.41	99.06	99.23	99.86	97.53	99.59	**100**	99.82
C7	85.41	83.71	83.40	81.95	77.42	85.33	**92.20**	83.42	77.55	87.88	88.98
C8	91.05	93.36	93.81	91.90	93.18	92.01	93.89	93.13	94.76	**95.73**	95.69
C9	59.01	44.35	57.93	43.64	49.71	46.43	38.85	64.97	47.11	**72.31**	57.76
C10	82.09	65.28	72.88	68.09	70.67	69.20	29.81	76.19	75.95	77.91	**84.41**
C11	**96.87**	96.14	95.74	90.36	93.41	96.14	95.82	95.10	95.50	93.97	96.25
OA(%)	82.32	84.70	84.24	80.63	81.47	84.01	83.85	84.70	82.35	85.23	**86.07**
AA(%)	82.14	80.68	82.23	77.74	79.57	80.43	78.72	82.71	80.03	83.34	**85.14**
$\kappa(\times 100)$	77.26	80.11	79.61	75.06	76.12	79.18	79.31	80.09	77.13	80.67	**81.95**

different data sources through interactive enhancement of multimodal features and achieves relatively excellent classification results on the three datasets.

The GLT model effectively leverages the complementary strengths of CNN and Transformer to extract both local and global semantic information from different modalities. This approach enables GLT to achieve comparable classification accuracy to S2ENet. In contrast, the MFT model, which directly utilizes LiDAR data as classification tokens, fails to achieve the expected performance, particularly on the

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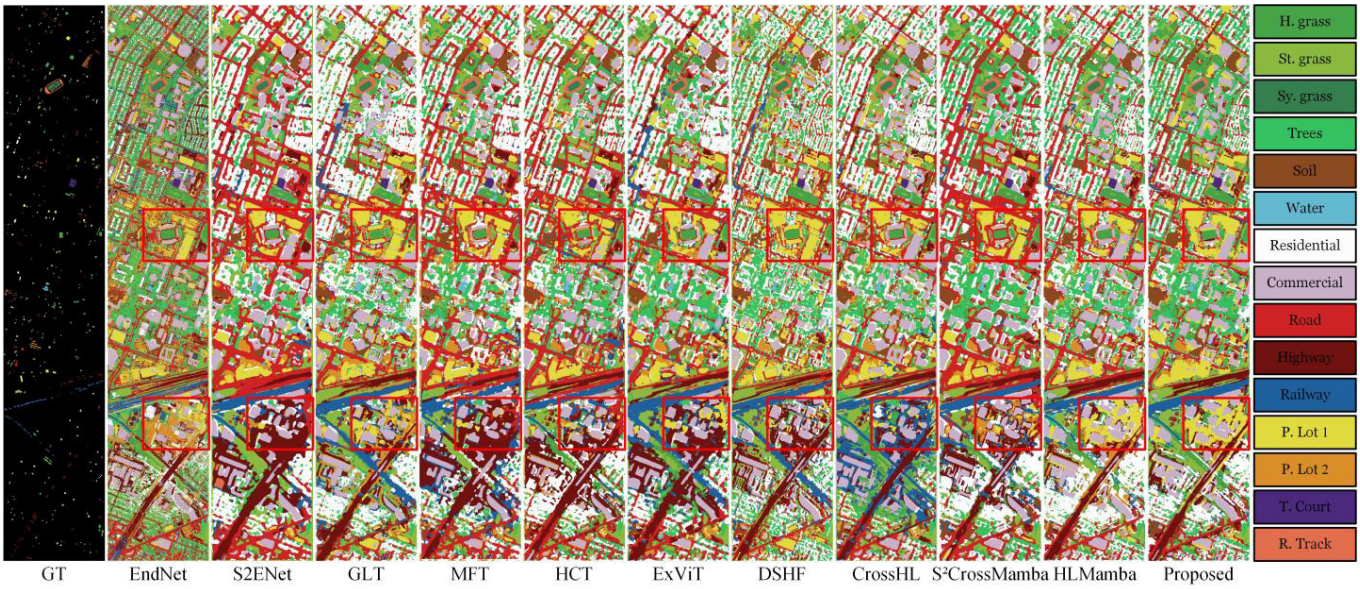


Fig. 12. Classification maps obtained by different methods on the Houston 2013 dataset.

MUUFL dataset, where it underperforms even EndNet. HCT, while demonstrating strong performance on the Houston 2013 dataset, exhibits limited generalization ability, as evidenced by its results on the Trento and MUUFL datasets.

ExViT enhances the performance of ViT by refining its architecture, leading to improved classification results across all three datasets. DSHF, by progressively enhancing both shallow and deep modal features, achieves strong performance on certain datasets, although its performance on Houston 2013 remains suboptimal. CrossHL effectively establishes long-range relationships between elevation features and spatial–spectral features through an improved multihead self-attention mechanism, resulting in better classification performance on all three datasets. S^2CrossMamba enhances the feature representation ability through a cross-SSM structure. However, the performance of this model is limited by the single scanning strategy. HLMamba, as an SOTA method based on Mamba, leverages Mamba to fully excavate and fuse spectral–spatial, elevation, and edge information, surpassing other comparative methods in terms of performance.

Although the above methods have demonstrated excellent classification accuracy, the proposed method has more impressive performance.

Specifically, on the Houston 2013 dataset, its OA, AA, and κ coefficient increase by 0.54%, 0.42%, and 0.58, respectively, compared to HLMamba. A further analysis of class-wise accuracy reveals that the proposed method achieves the highest classification accuracy in 7 out of 15 land cover categories. Notably, the accuracy improvements for C9 (roads) and C11 (railways) are particularly prominent, achieving 3.13% and 1.59% enhancements, respectively, compared to the best comparative methods such as HLMamba and ExViT.

Although the classification difficulty of the Trento dataset is relatively low, the proposed method still maintains certain advantages. Notably, it demonstrates leadership in four out of six categories, with classification performance significantly superior to comparative methods. On the more complex MUUFL dataset, the proposed method still maintains excellent performance. Compared with HLMamba, the proposed method achieves improvements of 0.84%, 1.80%, and 1.28 in OA, AA, and κ coefficient, respectively. This result further corroborates the effectiveness and robustness of the proposed method.

The experimental results demonstrate that the proposed method significantly outperforms SOTA methods in classification performance. Particularly, by introducing a bidirectional sequence scanning mechanism, this approach surpasses S^2CrossMamba and HLMamba—both supporting unidirectional scanning only—in global contextual modeling capabilities. Such bidirectional interaction mechanisms effectively enhance the model's ability to capture long-range dependencies, thereby improving the accuracy of classification tasks.

H. Classification Maps

Figs. 12–14 show the classification maps of the proposed method and some comparison methods on three datasets. On the Houston 2013 dataset, we selected two representative areas, both characterized by a mixture of "Parking Lot 1" and "Commercial." The experimental results reveal that several methods, including EndNet, GLT, and MFT, misclassify "Commercial" areas as "Parking Lot 1" in the first area. Similarly, in the second area, most methods erroneously classify "Parking Lot 1" as "Highway" or "Railway." It is worth noting that the HLMamba method using the Mamba framework has shown better performance in both areas, although there are also some minor classification errors; it accurately identifies the real ground object categories.

Despite the relatively small number of categories and the continuous distribution of objects on the Trento dataset, most comparison methods still produce noisy classification results. EndNet, for instance, exhibits significant salt-and-pepper noise in its predictions. S2ENet, HCT, and CrossHL struggle with category confusion in unlabeled regions. In contrast, methods like GLT, ExViT, and HLMamba demonstrate better visualization performance. The MUUFL dataset can better reflect the

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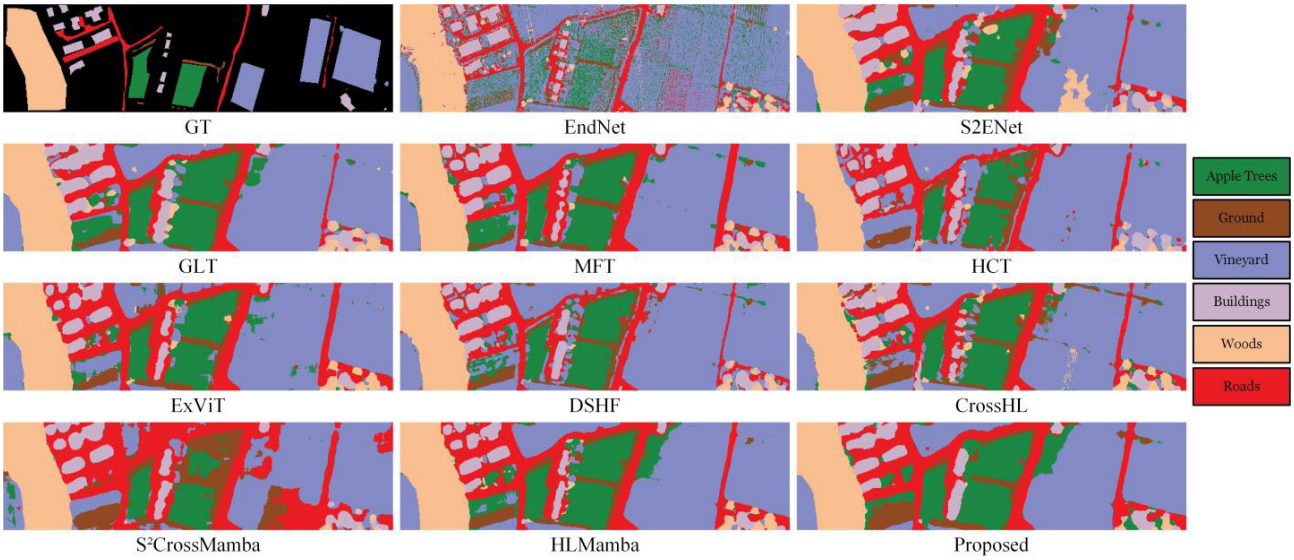


Fig. 13. Classification maps obtained by different methods on the Trento dataset.

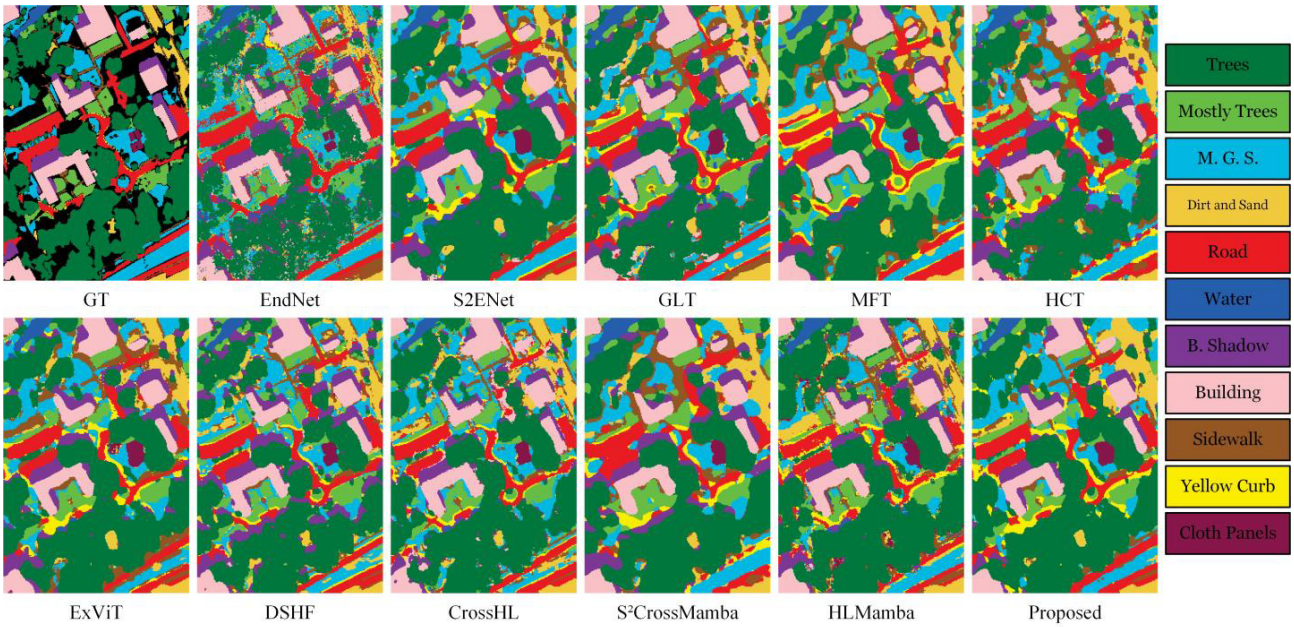


Fig. 14. Classification maps obtained by different methods on the MUUFL dataset.

performance of different methods under supervised learning due to its high coverage of labeled samples. The experimental results show that most comparison methods still have different degrees of "salt-and-pepper noise" problems on the MUUFL dataset.

Through the mutual guidance of multimodal information in the CMKS module, the proposed method can accurately identify complex objects in the Houston 2013 dataset. In addition, the proposed method also achieves better classification results than the comparative methods on the Trento and MUUFL datasets, and the visualization maps are more accurate.

I. Computing Complexity Analysis

This section conducts a computational complexity analysis of the proposed method against four SOTA methods (including two Mamba-based models), namely GLT, CrossHL, S^2CrossMamba, and HLMamba, from three indicators: inference time, floating-point operations (FLOPs), and parameters (Params).

The experimental results are shown in Fig. 15. Specifically, on the Houston 2013 dataset [Fig. 15(a)], the proposed method shows certain shortcomings in terms of inference time, FLOPs, and Params. This can be attributed to its "space-for-accuracy" strategy, which enhances modeling capabilities through multiscale large-size input feature maps.

On the Trento dataset [Fig. 15(b)], compared with Fig. 15(a), the proposed method achieves dual reductions in both inference time and parameter count while maintaining leading classification accuracy.

On the MUUFL dataset [Fig. 15(c)], it is evident that the proposed method significantly outperforms other methods in inference time. Compared with HLMamba, which adopts a

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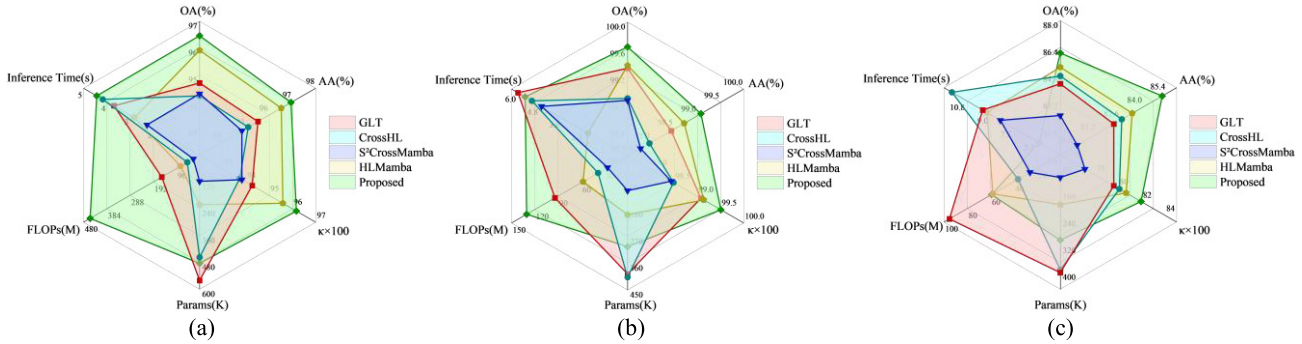


Fig. 15. Comparison between Mamba and Transformer. (a)–(c) Houston 2013, Trento, and MUUFL, respectively.

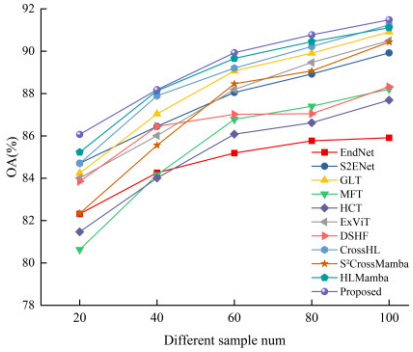


Fig. 16. Comparison of different training sample sizes on the MUUFL dataset.

similar multiscale strategy, the FLOPs consumption is comparable. Meanwhile, the parameter count also shows obvious advantages over GLT and CrossHL.

J. Different Number of Training Samples

To validate the robustness of the proposed method, a comparison experiment with training sample sizes ranging from 20 to 100 was conducted on the MUUFL dataset. As shown in Fig. 16, the classification performance of all methods exhibits an upward trend with the increase in sample size. However, the proposed method consistently maintains performance advantages across the entire sample size range, especially in scenarios with extremely limited training data. This experimental result fully corroborates the robustness of the proposed method.

K. T-SNE Feature Visualization

This section employs the t-SNE algorithm [50] to perform dimensionality reduction visualization and comparative analysis of the feature representations from the proposed method, the original dataset, and two Mamba-based models (S^2CrossMamba and HLMamba). The experimental results are shown in Fig. 17, with specific analysis as follows.

On the Houston 2013 dataset, the feature distribution of the original dataset exhibits significant mixing, with unclear boundaries between class features. In contrast, although S^2CrossMamba and HLMamba show improved feature separability, obvious interclass overlap remains. The proposed method effectively enlarges interclass distances while compressing intraclass distances through the bidirectional scanning

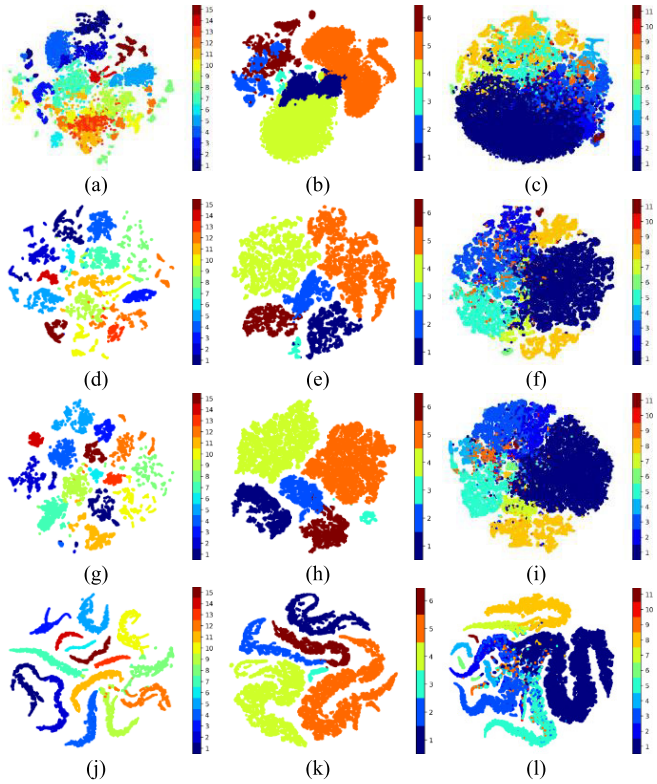


Fig. 17. t-SNE feature visualization. (a)–(c) Origin Houston 2013, Trento, and MUUFL, respectively. (d)–(f) S^2CrossMamba. (g)–(i) HLMamba. (j)–(l) Proposed.

mechanism, achieving discriminative optimization of the feature space. Especially for classes C8 (Commercial) and C12 (Parking Lot 1), the visualization demonstrates stronger class distinguishability.

On the Trento dataset, heterogeneous feature mixing still exists in the visualization of original features. S^2CrossMamba and HLMamba exhibit local feature overlap in classes C2 (Buildings) and C6 (Roads).

On the MUUFL dataset, as a highly complex dataset, the original features show no class separation, significantly increasing visualization difficulty. Although the proposed method has slight confusion in a few marginal classes, the degree of confusion is far lower than that of S^2CrossMamba and HLMamba, highlighting its strong robustness.

The visualization results indicate that the proposed method achieves dual optimization of the feature space through

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multiscale feature extraction and bidirectional sequence scanning strategies.

V. Conclusion

To overcome the limitations of existing Mamba-based joint classification methods in dealing with the noncausal characteristics of image data, this article proposes a novel TBi-Mamba method for the joint classification of HSI and LiDAR data. By introducing the CMKS module, this method effectively captures the local features between different modalities. Then, the TBi-M module models the global features of multimodality from the perspective of bidirectional sequence scanning. Finally, the MFRM module further improves the performance of the model through the auxiliary loss based on image reconstruction. The experimental results show that the proposed TBi-Mamba method can provide better classification performance than some SOTA methods on multiple datasets, verifying its effectiveness in multimodal remote sensing image classification tasks.

Notwithstanding, the proposed method still has certain limitations. On the one hand, the computational complexity of the proposed method is rather high in some scenarios; on the other hand, joint classification tasks still require more innovative scanning strategies to fully exploit the potential information in high-dimensional noncausal visual data. In future work, we will devote ourselves to addressing the above limitations in our research.

References

[1] C. M. Gevaert, J. Suomalainen, J. Tang, and L. Kooistra, "Generation of spectral–temporal response surfaces by combining multispectral satellite and hyperspectral UAV imagery for precision agriculture applications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 3140–3146, Jun. 2015.

[2] G. Zhan, Y. Uwamoto, and Y.-W. Chen, "HyperUNet for medical hyperspectral image segmentation on a choledochal database," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Las Vegas, NV, USA, Jan. 2022, pp. 1–5.

[3] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[4] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[5] R. Hänsch and O. Hellwich, "Fusion of multispectral LiDAR, hyperspectral, and RGB data for urban land cover classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 2, pp. 366–370, Feb. 2021.

[6] P. Ghamisi, B. Höfle, and X. X. Zhu, "Hyperspectral and LiDAR data fusion using extinction profiles and deep convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 6, pp. 3011–3024, Jun. 2017.

[7] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[8] M. Brell, K. Segl, L. Guanter, and B. Bookhagen, "Hyperspectral and LiDAR intensity data fusion: A framework for the rigorous correction of illumination, anisotropic effects, and cross calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2799–2810, May 2017.

[9] L. Fang, N. He, S. Li, P. Ghamisi, and J. A. Benediktsson, "Extinction profiles fusion for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1803–1815, Mar. 2018.

[10] C. Ge, Q. Du, W. Li, Y. Li, and W. Sun, "Hyperspectral and LiDAR data classification using kernel collaborative representation based residual fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1963–1973, Jun. 2019.

[11] W. Liao, A. Pižurica, R. Bellens, S. Gautama, and W. Philips, "Generalized graph-based fusion of hyperspectral and LiDAR data using morphological features," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 552–556, Mar. 2015.

[12] H. Chang et al., "Deep symmetric fusion transformer for multimodal remote sensing data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644115.

[13] H. Wu, Z. Du, D. Zhong, Y. Wang, and C. Tao, "FSVLM: A vision-language model for remote sensing farmland segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4402813.

[14] H. Wu, W. Mu, D. Zhong, Z. Du, H. Li, and C. Tao, "FarmSeg_VLM: A farmland remote sensing image segmentation method considering vision-language alignment," *ISPRS J. Photogramm. Remote Sens.*, vol. 225, pp. 423–439, Jul. 2025.

[15] Z. Kuang, H. Bi, F. Li, C. Xu, and J. Sun, "Polarimetry-inspired contrastive learning for class-imbalanced PolSAR image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5212819.

[16] H. Bi, H. Chang, X. Wang, and D. Hong, "Cross-attention-driven adaptive graph relational network for multilabel remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5224414.

[17] Z. Li, F. Liu, W. Yang, S. Peng, and J. Zhou, "A survey of convolutional neural networks: Analysis, applications, and prospects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 12, pp. 6999–7019, Dec. 2022.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. 31st Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 6000–6010.

[19] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," 2023, *arXiv:2312.00752*.

[20] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[21] Y. Liu et al., "VMamba: Visual state space model," 2024, *arXiv:2401.10166*.

[22] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[23] C. Shi, T. Wang, and L. Wang, "Branch feature fusion convolution network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5194–5210, 2020.

[24] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 764–773.

[25] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[26] S. Fang, K. Li, and Z. Li, "S^2ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[27] Y. Feng, L. Song, L. Wang, and X. Wang, "DSHFNet: Dynamic scale hierarchical fusion network based on multiattention for hyperspectral image and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5522514.

[28] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[29] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Montreal, QC, Canada, Oct. 2021, pp. 9992–10002.

[30] C. Shi, S. Yue, and L. Wang, "A dual-branch multiscale transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504520.

[31] X. Zhang et al., "A lightweight transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5517617.

[32] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[33] K. Ni, D. Wang, Z. Zheng, and P. Wang, "MHST: Multiscale head selection transformer for hyperspectral and LiDAR classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 5470–5483, 2024.

[34] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[35] S. K. Roy et al., "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

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[36] G. Zhao et al., "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.
[37] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[38] S. K. Roy, A. Sukul, A. Jamali, J. M. Haut, and P. Ghamisi, "Cross hyperspectral and LiDAR attention transformer: An extended self-attention for land use and land cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512815.
[39] J. Yao, D. Hong, C. Li, and J. Chanussot, "SpectralMamba: Efficient mamba for hyperspectral image classification," 2024, *arXiv:2404.08489*.
[40] J. X. Yang, J. Zhou, J. Wang, H. Tian, and A. W. C. Liew, "HSIMamba: Hyperpsectral imaging efficient feature learning with bidirectional state space for classification," 2024, *arXiv:2404.00272*.
[41] D. Liao, Q. Wang, T. Lai, and H. Huang, "Joint classification of hyperspectral and LiDAR data based on mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530915.
[42] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multi-scale feature fusion state space model for multi-source remote sensing image classification," 2024, *arXiv:2408.14255*.
[43] G. Zhang, Z. Zhang, J. Deng, L. Bian, and C. Yang, "S^2CrossMamba: Spatial–spectral cross-mamba for multimodal remote sensing image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.
[44] Y. He et al., "Classification of multisource remote sensing data using slice mamba," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5505414.
[45] Y. Li, D. Li, W. Xie, J. Ma, S. He, and L. Fang, "Semi-mamba: Mamba-driven semi-supervised multimodal remote sensing feature classification," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Apr. 15, 2025, doi: 10.1109/TCSVT.2025.3560615.
[46] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.
[47] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[48] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL gulfport hyperspectral and LiDAR airborne data set," Dept. Electr. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, Oct. 2013.
[49] X. Du and A. Zare, "Technical report: Scene label ground truth map for MUUFL gulfport data set," Dept. Electr. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep. 20170417, Apr. 2017.
[50] L. van der Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Nov. 2008.



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